

Notes for NGA LTER Physics Team Sampling

Version November 2025

See Ocean Best Practices group protocols!

<https://www.oceanbestpractices.org/ocean-best-practices-systems/repository/>

Bring to Seward:

- Blank hard drive or thumb drive to bring data home
- PC computer with SUNAcom program
- Camera (and take lots of photos!)
- Latest version of the Hydrolog Excel file for logging cast info.
- 3-ring binder to store Hydrolog sheets
- PPE, dry bag, etc from O'Neill 103 oceanography shop if needed

Get from Mooring Loft in Seward:

- Tool Box with tools for expected needs (e.g., SUNA hose clamp driver)
- Kim Wipes (For connectors, not for SUNA lens)
- Lens paper (for SUNA lens)
- Isopropyl Alcohol for cleaning SUNA/ISUS lenses
- Black electrical tape for SUNA mounting & misc other things
- Silicone Grease
- Parafilm
- SUNA (may be in warehouse), SUNA battery & spare, cables, charger, extension cord

If doing Mooring Recovery, bring from Mooring Loft:

- Deck box and dunk transducer
- Scrubbing materials for cleaning recovered gear
- Tires for the number of large floats to be recovered
- Boxes for instruments
- Computers with software and connection cables for downloading instruments
- Deployment sheet with latitude/longitude and release codes for the mooring to be recovered
- Tool box

If doing Mooring Deployment, bring from Mooring Loft:

- Electrical tape, kim wipes, silicone grease
- Instruments, anchors, etc to be deployed
- Computers with software for communicating with the instruments
- Deployment sheets to be filled in with serial numbers and release codes
- Deck box and dunk transducer
- Target deployment locations
- Tool box

Prior to Cruise

- Prep for image support: set up S2S (Ship to Shore) folder for SSH, SST, Chl_a daily files
 - Check with Rachel

Event Logger:

- Is accessible from any laptop connected to the ship server, but typically there will be an LTER-supplied designated laptop in the electronics/computer room on the ship, that will have the CTD event logger running during the cruise.
- Record every time the CTD goes into the water and every time it comes out of the water.
- Record all mooring deployments and recoveries
- Record any instances of other instrumentation fixes or discrete samples that you take
- Record anything out of the ordinary for which you want a time/position fix
- Clean up entries before the end of the cruise (you can add missed events, edit old events, etc.)

CTD

- Ship-supplied sensors
 - Temperature x2, Conductivity x2, Oxygen x2, Fluorometer, Transmissometer, PAR, Altimeter
- Science-supplied instruments
 - SUNA (physics team), LISST (particle team), UVP (particle team)
 - SUNA: work with Marine Technicians (MarTechs) to install SUNA at start of cruise
 - Make sure you have the SUNA, battery, hookup cables between CTD/SUNA/batter, battery charger cable, extension cord for charger, computer cable, and power supply, isopropyl alcohol, lens paper (**not kimwipes**), parafilm
 - Make sure the battery is charged before sailing.
 - Computer with SUNAcom program
 - Before start of the cruise, connect to the SUNA with computer. USB connection works for the transfer of data/settings check. A functionality test while also maintaining a connection to the computer requires more power, see below.
 - Note number of files on the flash drive
 - Check internal clock on SUNA and update if necessary
 - Do a rough functionality check with DIW and nutrient standard to check offsets and slope. The SUNA will not collect a measurement unless it is either a) powered through the black and red alligator clips on the communication cable via an external power supply that the MarTech can provide. **Do not supply over 13 V to the SUNA.**
 - Feel free to do a careful recalibration (requires DIW). SUNA manual has more detailed steps.
 - During the cruise
 - Recharge the battery each night. Clean contacts if they get a bit green. Avoid dumping/splashing water onto the charger and its power cord.
 - It is good to periodically clean the windows with lens paper and isopropyl alcohol.
 - Avoid re-calibrating in the middle of the cruise
 - It is good to download data on occasion and check the spectra for signs of large changes that may indicate the need for servicing of the instrument after the cruise.
 - At end of cruise
 - Connect to SUNA (power supply not needed)
 - Download all raw data files from this cruise onto your computer.
 - Add this data to the cruise hard drive that the MarTech will provide to the chief scientist
- CTD Hydrographic Profile Log Sheets

- o Print out Hydrolog sheets from the Excel file – one for each CTD cast (about 50-80 for the cruise) and log the header information at the top of each sheet.
- o Fill in the nominal target trip depths when provided by the phytoplankton team
- o Fill in the actual pressure depth of the bottle trips as each bottle is fired
- o No need to fill in the sample columns unless another team requests this.
- o Fill in anything unusual about the cast, data, data quality in the comments section or anywhere else on the log sheet.
- o Put each log sheet in a 3-ring binder and bring back to Fairbanks with the data
- Making casts
 - o File naming convention: **SKQ2021yyS_xxx.hex** where xxx is Consecutive Station # beginning with 001 and yy is a consecutive cruise number provide by the ship.
 - o Every time the CTD goes into the water the Consecutive Station number is incremented
 - o Soak CTD for 60 seconds after pump comes on. If weather is good you can soak at the surface. If in wavy conditions, soak at 10 m depth. If some instrument requires a deeper soak then adjust per user request.
 - o Do not collect data during the soak.
 - o Bring CTD back to surface following soak.
 - o **Turn on data collection at the start of the downcast.**
 - o Keep an eagle eye on the altimeter and **avoid ever touching down.**
 - o Try to get a CTD bottom sample within 5 m of the bottom for all stations on the shelf, but beware strong currents and variable topography near Middleton Island and coastal regions that might dictate extra caution and more conservative operations.
 - o For stations > 500 m depth, 10 m off the bottom is OK.
 - o Wait 60 seconds before tripping bottles at each depth level stop.
 - o At the end of each cast, run command-line scripts on the CTD computer to generate generic bottle files, 1-m average data files and sound velocity profiles.

Underway Data

RV Sikuliaq:

- Data are stored under `.../cruisename/lids/...`
- After 4 pm, the SKQ MarTech can create a “daily average file” of the prior day’s worth of data.
- The daily average files can be run through the MapUnderway.m script
- The MapUnderway.m script requires you to individually download the ISUS data from `.../cruisename/lids/raw/mbari_isus_fwd` if you want to plot the NO3 data
- Monitor the underway data streams through the cruise, report data quality issues to the MarTech
- ISUS
 - o Before the cruise, it is good to work with the MarTech too clean the optics (isopropyl alcohol, lens wipe) and do a functionality test with DIW, standard, and/or USW water.

RV Tiglax:

TSG

- Only thermosalinograph (TSG) can be set up for the flow through system on this ship. The TSG used is an SBE 21.
- The SBE 21 should already be hooked up in the engine room and the data is transmitted from there to the interface box in the electronics room (back room of the bridge). Be aware that the power cable for the SBE 21 should be plugged into the I/O connector on the SBE 21,

and not the SBE 38 connector. The cable can fit into either, unfortunately. The R/V Tiglax crew knows this, but you should know it too.

- For a system check: (More details can be found in the SBE 21 user manual or SeaCAT and Sealogger Navigation Interface Box user manual via the Seabird Scientific website).
 - Connect the interface box to a computer via a serial-usb connection. Be sure to verify COM ports, use the Device Manager program in Windows to assist with that. This computer should have Seasoft installed.
 - Open up Seaterm (not SeatermV2) and select the “SBE 21” option.
 - Turn on the interface box using the toggle switch, that action should bring up a response in Seaterm about the interface box.
 - In Seaterm click the “Connect” button. You should see *ds followed by some values the last of which is probably ‘N’. The ‘N’ means not logging.
 - Enter the command ‘GL’. You should now see hexadecimal values in the terminal every 10 seconds. This means the SBE 21 is now sending data through the interface box to the computer.
 - Close Seaterm
 - Open Seasave and set up ‘Configure Inputs’ to use the SBE 21 xmlcon file.
 - NMEA data feed typically comes through a serial usb cable that connects directly to the computer, so you can doublecheck that this option is selected in the xmlcon file.
 - Save your changes, then go to Real-Time Data>Start...
 - Create a file to save data to and make sure you will be archiving data and click ‘Start’.
 - Set up display how ever you like.
 - **Ask the ship to turn on the pumps for the TSG!** Otherwise you’ll just record lukewarm water for the entire cruise.

MET station

- Usually this is located inside the warehouse. Big pole with a white box and a bunch of sensors for metrological data.
- Open the white box and check that there is a micro SD card in there.
- Bolt the MET station to the railing of the Tiglax flying bridge. See image below.
- Plug in the extension cord. The MET station is now powered and is collecting data.
- How can I tell if it’s collecting data? It’s not easy. There is a small LED light next to the microSD card but it’s not very informative.
 - The light will be a solid red color when the card is being read. This can happen if you insert a microSD card while the system is powered.
 - The light will be a solid orange color if the card is experiencing an error.
 - When you remove the card, you should click the eject button next to the light. The light will turn green and you have 15 seconds to remove the card. This is the best practice.
 - Unfortunately, there doesn’t seem to be a warm fuzzy green blinking option to show that data is actually being written. What follows is the work around.

To read data on the microSD, eject the microSD and upload the .dat file to a computer. The computer should have loggernet software downloaded to it (or you can get a 30 day free trial option from their website). The steps to import the data into loggernet and then export it in a readable format are as follows.

import and export the data are:

- 1) open up LNDB manager (that is part of the loggernet software package).
- 2) ignore pop ups about setting up LN and stuff.
- 3) Tools>import data.
- 4) Choose the .dat file from your directory. This is what you downloaded from the microsd card.
- 5) Under import info, select new table and call it whatever you want.
- 6) Click import, now the data is a table within loggernet software. You can read some but not all of the data while in the data review tab. To view all of the data proceed to the next step.
- 7) Tools>export data
- 8) In the "select database" pop up (if it pops up) just click OK.
- 9) In the database table drop down select the table you want to export.
- 10) In the export info I chose custom csv.
- 11) Select the date time range and click export.
- 12) You can go that file and now see that all the data is in fact there.

- 1) open up LoggerNet.
- 2) Navigate to Data > CardConvert
- 3) Click the "Select Card Drive" button. Choose the location to which you've copied the *.dat binary file from the MicroSD card.
- 4) Click the "Change Output Directory" button. Choose where you want the converted *.dat ascii file to go.
- 5) Click the "Destination File Options" button. Change any settings as desired, but the defaults are fine for just checking the data.
- 6) Click the "Start Conversion" button. A notepad readable ASCII file is now in your desired folder!
- 7) Check that data has recorded (scroll to the bottom of the file if there are other deployments/cruises on the same card). Note UTC time!

Acoustics

- EK80, ADCP and multibeam systems will generally be operated by the MarTechs, but please keep an eye on them each day to ensure data collection is progressing well.
- Try to keep the multibeam, EK80 and ADCP mapping all going for bottom depths of up to 500-1000 m
- For depths > 500-1000 m, keep the EK80 running at night so we get backscatter data along with the net tows.
- For depths > 1000 m we can go with either the multibeam or the EK80. Generally, we favor the multibeam at this time of day.
- Try to "fill in" gaps in seafloor mapping by selecting vessel routes on transits that do not exactly reproduce prior tracklines. Work the chief scientist to help inform the bridge of good routes.
- **ADCP**
 - Record data from the 75 KHz Ocean Surveyor (hull mount)
 - Typically in nb mode with 8 m bins
 - Record data from the 300 KHz Workhorse (drop keel mount)
 - Typically in nb mode with 2 m bins
 - Monitor the data streams through the cruise
 - Compare data quality and signals from the two instruments on the E-lab screen (other plots in the data files)
 - report data quality issues (especially stopped pinging!) to the MarTech

- If you have time, download and plot the data yourself. Matlab files are in:
.../cruisename/adcp/raw/cruisename/proc/instrumentname/vector/*.mat

Acrobat

- The Acrobat system typically gets set up in the aft port side of the E-lab as shown in the photo below.
- You can use some string to tie the deck box and monitors into the corner. Tie-down hardware is in the Main Lab (drawer on Starboard side).
- The deck cable to the winch gets strung out through the port above the blue rack (with the SBE deck boxes) into the hallway, along the wire chase in the hallway, through the two bulkheads up high, along the house and then over to the winch. The MarTechs can help you set up a line to make the up-in-the-air run across the deck. Start by hooking up in the Elab and then work your ways out to the winch. On deck, keep, attach the wire to the railing, and go from railing to the red crane. This keeps the wire off the deck itself, which is what the MarTech prefers. Ask questions if you are unsure!
- SKQ can provide electric power to the blue winch.
- I'd shoot for 6.5 to 7 knots **through the water** in good conditions (e.g. have the bridge try for 6.5 or 6 ¾ knots. Or a bit less in lots of waves. Be clear in comms with the bridge about target speed in terms of speed through water vs. speed over ground.
- For ACROBAT to winch setup, always avoid sharp 90 degree bends in the cable.

Near end of cruise:

- Clean up Event Logs

At end of cruise:

- Return all tools, computers, SUNA supplies etc to Mooring Loft
- Return mooring gear to Mooring Loft
- Bring all data back to Fairbanks on a hard drive or thumb drive
- Bring back the log sheet binder to Fairbanks.
- Return any borrowed PPE/Drybags/etc.

As *Alpha Helix* Captain Bill Rook used to say: we can replace equipment but not people.

We can't replace lost data but don't damage the people for the sake of data.

Have fun!

Be safe!



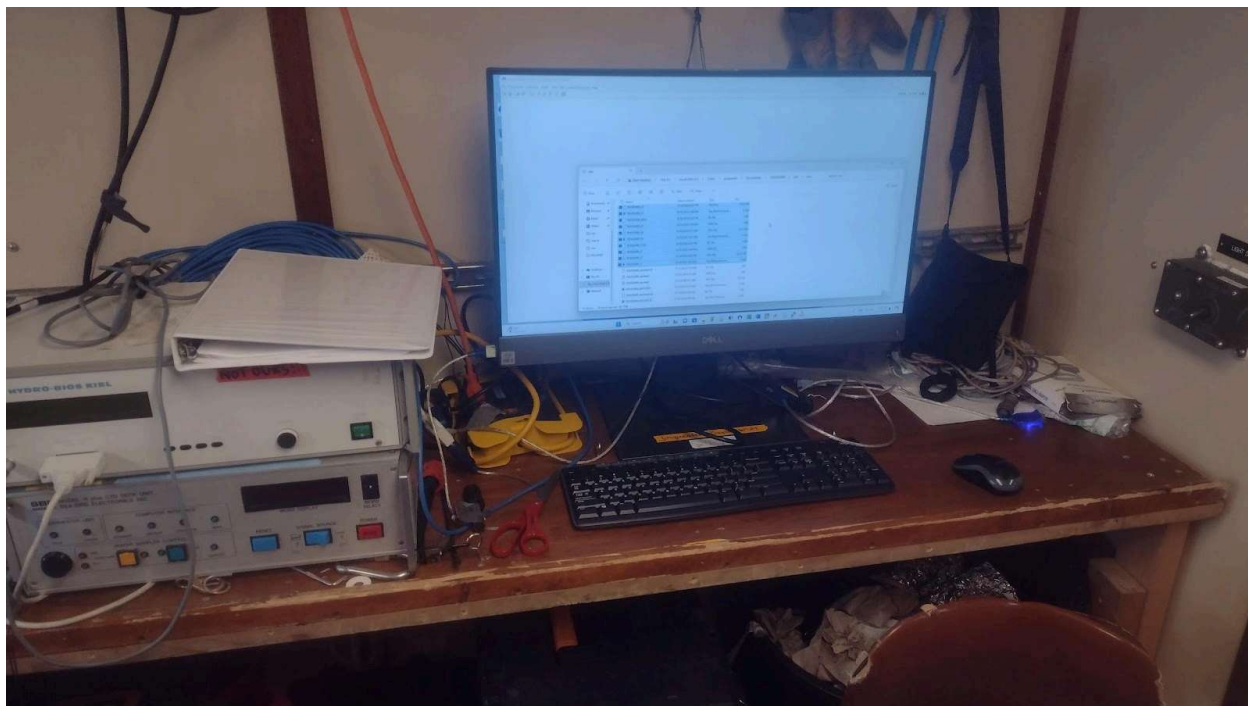
SUNA setup on Sikuliaq.



Acrobat setup on Sikuliaq.



SUNA setup on Tiglax



CTD deckbox and computer set up on Tiglax.



MET station for the TIGLAX set up.